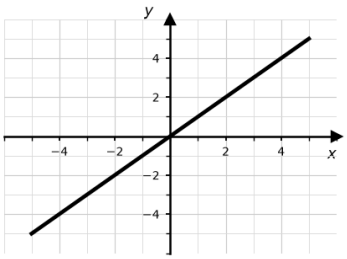
Algebra — Objective 4-1
Identifying Linear Functions

TEACHER ANSWER KEY — answers in red

Part 1 — Identify from a Graph

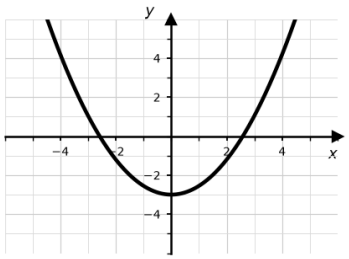
1.



Linear / Not Linear

How do you know? **straight line**

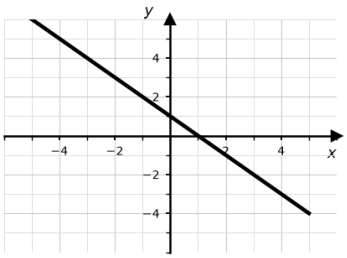
2.



Linear / **Not Linear**

How do you know? **curve (parabola)**

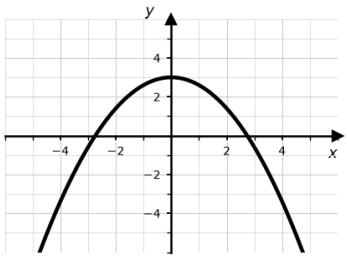
3.



Linear / Not Linear

How do you know? **straight line**

4.



Linear / **Not Linear**

How do you know? **curve (parabola)**

Part 2 — Identify from an X-Y Table

5.

x	y
-2	-5
-1	-2
0	1
1	4

Linear / Not Linear

How do you know? **y goes up +3 each step (constant rate)**

6.

x	y
-2	4
-1	1
0	0
1	1

Linear / **Not Linear**

How do you know? **y-change not constant (-3, -1, +1)**

7.

x	y
0	7
1	5
2	3
3	1

Linear / Not Linear

How do you know? **y goes down -2 each step (constant rate)**

8.

x	y
1	1
2	4
3	9
4	16

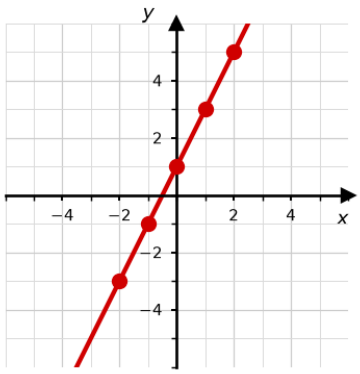
Linear / **Not Linear**

How do you know? **y-change not constant (+3, +5, +7)**

Part 3 — Find Y and Graph

9. y = 2x + 1

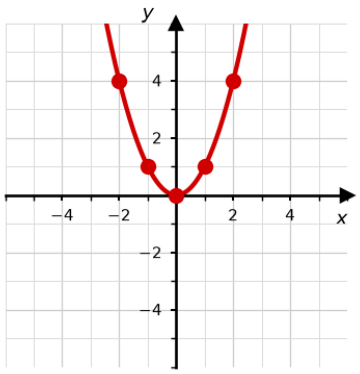
x	y
-2	**-3**
-1	**-1**
0	**1**
1	**3**
2	**5**



Linear / Not Linear

10. y = x²

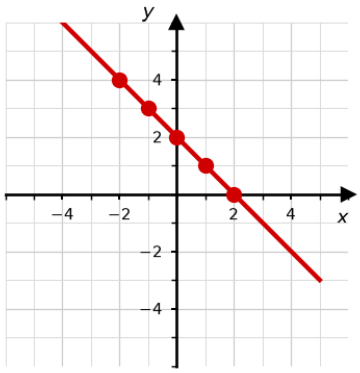
x	y
-2	**4**
-1	**1**
0	**0**
1	**1**
2	**4**



Linear / **Not Linear**

11. y = -x + 2

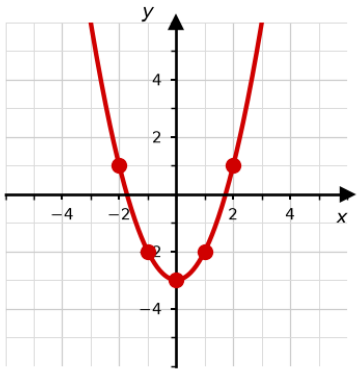
x	y
-2	**4**
-1	**3**
0	**2**
1	**1**
2	**0**



Linear / Not Linear

12. y = x² - 3

x	y
-2	**1**
-1	**-2**
0	**-3**
1	**-2**
2	**1**



Linear / **Not Linear**